Jing Liu 刘璟

Senior Researcher
Systems Research Group
Microsoft Research Asia

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Research Interests

I am a computer systems researcher specializing in operating systems, storage systems, and distributed systems. My current research particularly focuses on the CPU-side systems stack for agentic AI and the co-evolution of next-generation computing systems and models. I pursue holistic, full-stack designs in which in-situ code generation can eliminate cross-layer overheads and indirections to attain maximal end-to-end efficiency.

Education

Department of Computer Sciences, University of Wisconsin – Madison
Ph.D. in Computer Sciences — Aug. 2024
M.S. in Computer Sciences — May 2022
Advisors: Prof. Andrea Arpaci-Dusseau & Prof. Remzi Arpaci-Dusseau
Thesis: *Scale, Performance, and Fault Tolerance in a Filesystem Semi-Microkernel*

Software Institute, Nanjing University
B.E. in Software Engineering Rank: 1/238 — May 2016
Thesis: *Depth-aware layered edge for elastic RGB-D object proposal*

Professional Experience

Microsoft Research Asia — Oct. 2024 – Present
Senior Researcher.

Google, Madison, WI — Summer 2019
Software Engineering Intern. Mentor: Dr. Marc De Kruijf

Microsoft Research, Redmond, WA — Summer 2018
Research Intern. Mentor: Dr. Irene Zhang

Honors and Awards

Erik Riedel Best Paper Award, USENIX Conference on File and Storage Technologies (FAST) — 2025
Selected as one of 2 best papers from 167 submissions; top 1.2%

Meta PhD Research Fellowship, Meta Inc. — 2022 – 2024
Sole awardee in the distributed systems research area worldwide

Computer Sciences Department Golden Brick Award, UW-Madison — 2022

CS Summer Research Award, UW-Madison — 2017

Graduate Research Fellowship, UW-Madison — 2016 – 2017

Outstanding Undergraduate Thesis, University Highest Honor, Jiangsu Provincial Award — 2016

National Scholarship, Ministry of Education of China — 2013

Publications

† Student mentored by me.

Refereed Conference Publications

[1] Shawn (Wanxiang) Zhong, Junxuan Liao, **Jing Liu**, Mai Zheng, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Don't Let AI Agents YOLO Your Files: Information and Control in Agent-Native Filesystems*. In Proceedings of the 32nd ACM Symposium on Operating Systems Principles (SOSP '26), Prague, Czechia, September 2026. (arXiv:2604.13536)

[2] Yaoqi Chen, Jinkai Zhang, Baotong Lu, Qianxi Zhang, Chengruidong Zhang, **Jing Liu**, Jingjia Luo, Di Liu, Huiqiang Jiang, Qi Chen, Bailu Ding, Xiao Yan, Jiawei Jiang, Chen Chen, Mingxing Zhang, Yuqing Yang, Fan Yang, and Mao Yang. *RetroInfer: A Vector-Storage Approach for Scalable Long-Context LLM Inference*. In Proceedings of the 52nd International Conference on Very Large Data Bases (VLDB '26), August 2026. **(Covered by Hugging Face Trending Papers [url] and AI Research Roundup [url])**

[3] Sukjoon Oh, Minki Kang, Dohyun Kim, Baotong Lu, **Jing Liu**, Qianxi Zhang, Qi Chen, and Youjip Won. *Aker: Density-Aware Approximate Caching for Vector Search*. In Proceedings of the 52nd International Conference on Very Large Data Bases (VLDB '26), August 2026.

[4] Zhongjie Chen†, Wentao Zhang, Yulong Tang†, Ran Shu, Fengyuan Ren, Tianyin Xu, and **Jing Liu**. *Xkernel: Principled Performance Tunability of Operating System Kernels*. In Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI '26), July 2026.

[5] Chuandong Li, Ran Yi, Zonghao Zhang, **Jing Liu**, Changwoo Min, Jie Zhang, Yingwei Luo, Xiaolin Wang, Zhenlin Wang, and Diyu Zhou. *Aeolia: A Fast and Secure Userspace Interrupt-Based Storage Stack*. In Proceedings of the 31st ACM Symposium on Operating Systems Principles (SOSP '25), pages 479–495. ACM, October 2025.

[6] Shawn (Wanxiang) Zhong†, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Revealing the Unstable Foundations of eBPF-Based Kernel Extensions*. In Proceedings of the 20th European Conference on Computer Systems (EuroSys '25), pages 21–41. ACM, March 2025.

[7] **Jing Liu**, Yifan Dai, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Fast, Transparent Filesystem Microkernel Recovery with Ananke*. In Proceedings of the 23rd USENIX Conference on File and Storage Technologies (FAST '25), pages 1–18. USENIX Association, February 2025. **(Erik Riedel Best Paper Award)**

[8] Yifan Dai, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Symbiosis: The Art of Application and Kernel Cache Cooperation*. In Proceedings of the 22nd USENIX Conference on File and Storage Technologies (FAST '24), pages 51–69. USENIX Association, February 2024.

[9] **Jing Liu**, Anthony Rebello, Yifan Dai, Chenhao Ye, Sudarsun Kannan, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Scale and Performance in a Filesystem Semi-Microkernel*. In Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP '21), pages 819–835. ACM, October 2021.

[10] Irene Zhang, Amanda Raybuck, Pratyush Patel, Kirk Olynyk, Jacob Nelson, Omar S Navarro Leija, Ashlie Martinez, **Jing Liu**, Anna Kornfeld Simpson, Sujay Jayakar, et al. *The Demikernel Datapath OS Architecture for Microsecond-scale Datacenter Systems*. In Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP '21), pages 195–211. ACM, October 2021.

[11] Suli Yang, **Jing Liu**, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Principled Schedulability Analysis for Distributed Storage Systems using Thread Architecture Models*. In Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI '18), pages 161–176. USENIX Association, October 2018.

[12] Ramnatthan Alagappan, Aishwarya Ganesan, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Fault-Tolerance, Fast and Slow: Exploiting Failure Asynchrony in Distributed Systems*. In Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI '18), pages 390–408. USENIX Association, October 2018.

[13] **Jing Liu**, Tongwei Ren, Bingkun Bao, and Jia Bei. *Depth-aware Layered Edge for Object Proposal*. In Proceedings of the IEEE International Conference on Multimedia and Expo (ICME '16), pages 1–6. IEEE, July 2016.

[14] **Jing Liu**, Tongwei Ren, and Jia Bei. *Elastic Edge Boxes for Object Proposal on RGB-D images*. In Proceedings of International Conference on Multimedia Modeling (MMM '16), Lecture Notes in Computer Science, pages 199–211. Springer International Publishing, 2016.

arXiv Preprints

[1] Zihan Zhao†, Baotong Lu, Shengjie Lin†, Yizou Chen, **Jing Liu**, Yanqi Zhang, Ziming Miao, Ming-Chang Yang, Haiying Shen, Qi Chen, and Fan Yang. *Unifying Sparse Attention with Hierarchical Memory for Scalable Long-Context LLM Serving*. arXiv Preprint, 2026. (arXiv:2604.26837)

[2] Yuming Xu, Qianxi Zhang, Qi Chen, Baotong Lu, Menghao Li, Philip Adams, Mingqin Li, Zengzhong Li, **Jing Liu**, Cheng Li, and Fan Yang. *Scalable Distributed Vector Search via Accuracy Preserving Index Construction*. arXiv Preprint, 2025. (arXiv:2512.17264) (**Covered by Microsoft Research Focus [url]**)

Refereed Journals and Workshops

[1] Inho Choi, Anand Bonde, **Jing Liu**, Joshua Fried, Irene Zhang, and Jialin Li. *ML-native Dataplane Operating Systems*. In Proceedings of the 16th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '25), pages 8–14. ACM, October 2025.

[2] **Jing Liu**, Xiangpeng Hao, Andrea Arpaci-Dusseau, Remzi Arpaci-Dusseau, and Tej Chajed. *Shadow Filesystems: Recovering from Filesystem Runtime Errors via Robust Alternative Execution*. In Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '24), pages 15–22. ACM, 2024.

[3] Suli Yang, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Principled Schedulability Analysis for Distributed Storage Systems Using Thread Architecture Models*. ACM Transactions on Storage, 19(2):1–47, May 2023.

[4] **Jing Liu**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau, and Sudarsun Kannan. *File Systems as Processes*. In 11th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage '19). USENIX Association, July 2019.

[5] Irene Zhang, **Jing Liu**, Amanda Austin, Michael Lowell Roberts, and Anirudh Badam. *I'm Not Dead Yet! The Role of the Operating System in a Kernel-Bypass Era*. In Proceedings of the Workshop on Hot Topics in Operating Systems (HOTOS '19), pages 73–80. ACM, May 2019.

[6] **Jing Liu**, Tongwei Ren, Yuantian Wang, Sheng-Hua Zhong, Jia Bei, and Shengchao Chen. *Object proposal on RGB-D images via elastic edge boxes*. Neurocomputing, 236:134–146, May 2017.

Service

Shadow PC Co-chair, EuroSys '26

Program Committee: SOSP '25, APSys '25, NSDI '26, OSDI '26, SOSP '26, FAST '27, HOTOS '27

External Review Committee: ATC '24

Artifact Evaluation Committee: ATC/OSDI '22

Topic (OS & Emerging Hardware) Moderator, SOSP '21 Diversity Workshop

President, W-ACM, UW Madison Chapter of ACM's Women in Computing — Aug. 2021 – Aug. 2022

Presentations and Invited Talks

Systems Research: What We Build, Why We Build, and How We Do It?

Invited talk at MSRA Interns Event, Beijing, China. — Aug. 2025

Invited talk at MSRA – PKU Open Day, Beijing, China. — Jul. 2025

Invited talk at MSRA Open Day to Students, Beijing, China. — Jun. 2025

Fast, Transparent Filesystem Microkernel Recovery with Ananke

Talk at FAST '25, Santa Clara, CA. — Feb. 2025

Scale and Performance in a Filesystem Semi-Microkernel.

Invited talk at UIUC, Champaign, IL. — Oct. 2022

Invited talk at ENS de Lyon – France, virtual. — Mar. 2022

Invited talk at Samsung MSL Tech Talk, virtual. — Jun. 2021

Talk at SOSP '21, virtual. — Oct. 2021

File Systems as Processes.

Talk at HotStorage '19, Renton, WA. — Jul. 2019

Elastic Edge Boxes for Object Proposal on RGB-D Images.
Talk at MMM '16, Miami, MI. Jan. 2016

Teaching Experience

Computer Sciences Department of UW–Madison, WI Summer 2023
Instructor
Class: Introduction to Operating Systems.

Computer Sciences Department of UW–Madison, WI Spring 2022, Spring 2023
Teaching Assistant
Class: Distributed Systems. Instructor: Prof. Andrea Arpaci-Dusseau

Computer Sciences Department of UW–Madison, WI Spring 2018
Teaching Assistant
Class: Advanced Operating System. Instructor: Prof. Andrea Arpaci-Dusseau

Software Institute of Nanjing University, Nanjing, China Spring 2015, Spring 2016
Teaching Assistant, and Head Teaching Assistant
Class: Computer Operating System. Instructors: Prof. Bin Luo & Prof. Jidong Ge

Mentoring

A General Inference Engine for Sparse Attention
Zihan Zhao, University of Virginia, Ph.D. student May 2025 – Aug. 2026
Shengjie Lin, Georgia Tech University, Master student Aug. 2025 – Aug. 2026

Principled Performance Tunability in Operating System Kernels
Zhongjie Chen, Tsinghua University, Ph.D. student Apr. 2025 – July. 2026
Yulong Tang, Renmin University of China, Master student Sep. 2025 – May. 2026

Smart Caching in Vector Storage Systems
Sukjoon Oh, KAIST, Ph.D. student Oct. 2024 – Present

Unstable Foundations of eBPF Programs (*EuroSys '25*)
Shawn Zhong, UW–Madison, Ph.D. student Jun. 2024 – May 2025

DiskANN Case Study for Application Reaction to Filesystem Failure
Luozhong Zhou, UW–Madison, Undergraduate May 2023 – Nov. 2023